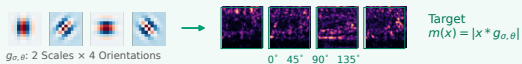


(a) Two Information Sources, Fused at Training Time Only

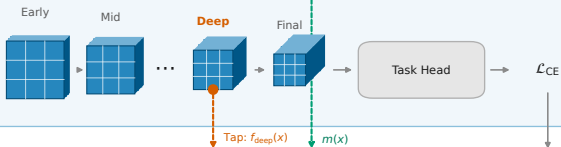


Input x

Source 1 · Hand-Crafted Knowledge: the Pinned Gabor Bank, Fixed Before Training



Source 2 · Learned Representation: the Backbone, Trained on the Cell's Own Data



Fusion · A Decaying Auxiliary Regression at the Tapped Stage

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda(t) \|Wf_{\text{deep}}(x) - m(x)\|_2^2$$

W : a 1x1 Convolution, Trained With the Network and Used Only Here

$\lambda(t)$: Cosine to Exactly 0,
Late Training Is Pure CE



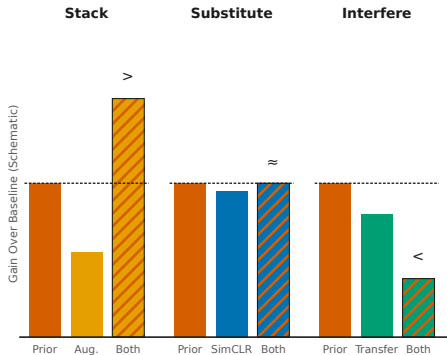
Deployment · W and the Bank Are Dropped: FLOPs and Parameters Identical to the Baseline



One Task Head, Trained for the Task at Hand;
the Same Bank, Tap and Schedule Serve All Three

(b) Fusion Outcomes: What Two Sources Can Do

Bars Schematic; Dashed Line: Better Single Source



Prior + DeiT Aug.:
+21.0 vs +13.3 Alone
(ViT, C100@10%)

Prior + SimCLR
≈ SimCLR Alone

Prior on ImageNet Init:
-16 at $\lambda_0 = 1.0$,
≈ 0 at $\lambda_0 \leq 0.3$

$$\Delta = G + \text{readout}$$

G : Frozen-Feature Gain (Linear Evaluation); readout: What the
Cell's Own Classifier Realizes